2022 SSMO Relay Round 1 Solutions

SMO Team

RR 1 Part 1: Suppose a, b, c are distinct digits where $a \neq 0$ such that $\left(\overline{abc}\right)^2 = \overline{bad00}$ where $d = a + b$. Find $a + 2b$.

Answer: $\boxed{16}$

Solution: Clearly, $c = 0$. So, we have

$$(100a + 10b)^2 = 10000b + 1000a + 100d \implies (10a + b)^2 = 101b + 11a.$$

The right hand side can be bounded by $101 \cdot 9 + 11 \cdot 9 = 1008$, meaning $10a + b \leq \sqrt{1008} \implies a \leq 3$. Testing, we have

$$a = 2 \implies b = 7 \implies a + 2b = 2 + 2 \cdot 7 = \boxed{16}.$$

RR 1 Part 2: Let $T = TNYWR$. Suppose lines ℓ and m have equations $y = (2+\sqrt{3})x+16$ and $y = \frac{\sqrt{3}}{3}x + 20$, respectively. Let A be a point on ℓ such that the shortest distance from A to m is T. Let O be the point on m such that $\overline{AO} \perp m$, and let P be the point on ℓ such that $PO \perp \ell$. Find PO^2.

Answer: $\boxed{128}$

Solution: Lines ℓ and m form an angle of

$$\arctan(2 + \sqrt{3}) - \arctan\left(\frac{\sqrt{3}}{3}\right) = 75^\circ - 30^\circ = 45^\circ.$$

So, APO is an isosceles right triangle with hypotenuse $AO = T = 16$, giving

$$PO = \frac{16}{\sqrt{2}} = 8\sqrt{2} \implies PO^2 = (8\sqrt{2})^2 = \boxed{128}.$$

RR 1 Part 3: Let $T = TNYWR$. Consider triangle ABC with $AB = T$, $AC = 100$, and $\angle ABC = 36^\circ$. Find the product of all possible values of BC.

Answer: $\boxed{6384}$

Solution: From the Law of Cosines, we have

$$\frac{AB^2 + BC^2 - AC^2}{2(AB)(BC)} = \cos 36^\circ \implies$$
$$\frac{T^2 + BC^2 - 100^2}{2(T)(BC)} = \cos 36^\circ \implies$$
$$BC^2 - BC(2T\cos 36^\circ) + (T^2 - 100^2) = 0.$$

From Vieta's formulas, the product of the two possible values of BC is

$$T^2 - 100^2 = 128^2 - 100^2 = \boxed{6384}.$$

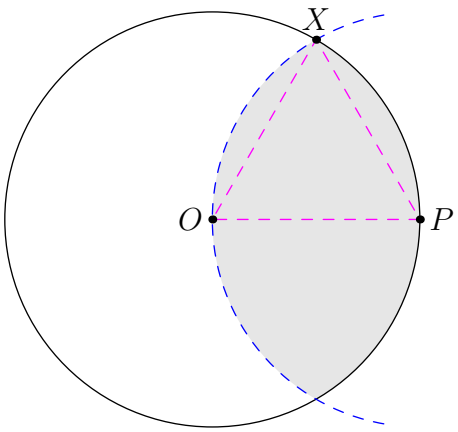
2022 SSMO Relay Round 2 Solutions

SMO Team

RR 2 Part 1: Let P be a randomly selected point on a circle, and let A be a randomly selected point inside the same circle. A dilation centered at P with a scale factor of 2 sends A to A'. The probability that PA' is less than the length of the diameter of the circle can be expressed as $\frac{a\pi - b\sqrt{c}}{d\pi}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a, b, d) = 1$. Find $a + b + c + d$.

Answer: $\boxed{16}$

Solution:



Assume WLOG that the circle is a unit circle and fix P. Draw a unit circle centered at P. For PA to be less than the length of the radius of the circle, A must lie in the shaded region. To compute the area of the shaded region, we divide it into two equal halves through line OP. Then, we can compute the area of each half by adding the areas of sectors OX and PX and subtracting the overlapping equilateral triangle POX of side length 1. This gives an area of

$$2\left(\frac{\pi}{6} + \frac{\pi}{6} - \frac{\sqrt{3}}{4}\right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

and a probability of

$$\frac{\frac{2\pi}{3} - \frac{\sqrt{3}}{2}}{\pi} = \frac{4\pi - 3\sqrt{3}}{6\pi} \implies 4 + 3 + 3 + 6 = \boxed{16}.$$

RR 2 Part 2: Let $T = TNYWR$. Suppose that the monic quadratic $f(x)$ is tangent to the function $g(x) = |x + 2| - T$ at two points. The value of $|f(1)|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{31}$

Solution: First, note that the parabola must be symmetric about the line $x = -2$, meaning it is of the form $(x+2)^2 + c$, for some constant c. The right side of $g(x)$ is represented by $(x+2) - T = x - 14$. For $f(x)$ to be tangent to $g(x)$, there must only be one solution to

$$(x+2)^2 + c = (x-14) \implies x^2 + 3x + (18+c) = 0.$$

Completing the square, we have

$$\left(x + \frac{3}{2}\right)^2 = \frac{9}{4} - (18+c).$$

Since there is only one solution, we must have

$$\frac{9}{4} - (18+c) = 0 \implies c = -\frac{63}{4}.$$

So,

$$|f(1)| = \left|(1+2)^2 - \frac{63}{4}\right| = \frac{27}{4} \implies 27 + 4 = \boxed{31}.$$

RR 2 Part 3: Let $T = TNYWR$. Suppose that

$$x + y = \left\lfloor \sqrt{T} \right\rfloor \quad \text{and}$$
$$x^5 + y^5 = 15.$$

The absolute difference between the two possible values of xy can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{77}$

Solution: We have $x + y = \left\lfloor \sqrt{T} \right\rfloor = \lfloor \sqrt{31} \rfloor = 5$. Let $xy = t$. Note that

$$\begin{aligned} x^2 + y^2 &= (x+y)^2 - 2xy = 25 - 2t \\ x^3 + y^3 &= (x+y)^3 - 3xy(x+y) = 125 - 15t \\ x^5 + y^5 &= (x^2+y^2)(x^3+y^3) - x^2y^2(x+y) \\ &= (25-2t)(125-15t) - 5t^2 \\ &= 25(t^2 - 25t + 125). \end{aligned}$$

So,

$$\begin{aligned} 25(t^2 - 25t + 125) = 15 &\implies \\ t^2 - 25t + \frac{622}{5} = 0 &\implies \\ t = \frac{25 \pm \sqrt{\frac{637}{5}}}{2}&. \end{aligned}$$

The absolute difference between the two possible values of $t = xy$ is

$$\frac{25 + \sqrt{\frac{637}{5}}}{2} - \frac{25 - \sqrt{\frac{637}{5}}}{2} = \sqrt{\frac{637}{5}} = \frac{7\sqrt{65}}{5} \implies 7 + 65 + 5 = \boxed{77}.$$

2022 SSMO Relay Round 3 Solutions

SMO Team

RR 3 Part 1: Let $f : \mathbb{Z} \to \mathbb{Z}$ be a function such that $f(0) = 0$ and $f(|x^2-4|) = 0$ whenever $f(x) = 0$. Suppose further that $|f(x+1) - f(x)| = 1$ for all $x \in \mathbb{Z}$. Let N be the number of possible sequences $\{f(1), f(2), \ldots, f(21)\}$. Find the remainder when N is divided by 1000.

Answer: $\boxed{40}$

Solution: Since $f(0) = 0$, we have $f(|0^2 - 4|) = f(4) = 0$ and $f(|4^2 - 4|) = f(12) = 0$. Define $\delta_x = f(x+1) - f(x)$, so $\delta_x \in \{-1, 1\}$ for all $x \in \mathbb{Z}$. Since $f(4) = 0$, we must have

$$\delta_0 + \delta_1 + \delta_2 + \delta_3 = 0,$$

so exactly two of $\{\delta_0, \delta_1, \delta_2, \delta_3\}$ are -1 and the other two are 1, giving $\binom{4}{2} = 6$ choices. Since $f(12) = 0$, we must have

$$\delta_4 + \delta_5 + \cdots + \delta_{11} = 0,$$

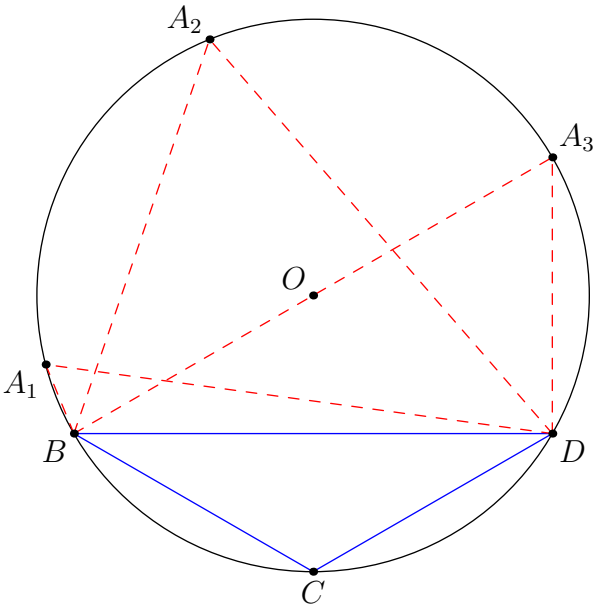
so exactly four of $\{\delta_4, \ldots, \delta_{11}\}$ are -1 and the other four are 1, giving $\binom{8}{4} = 70$ choices. Finally, each of $\delta_{12}, \ldots, \delta_{20}$ can be chosen freely from $\{-1, 1\}$, giving 2^9 choices. Thus the total number of sequences $\{f(1), f(2), \ldots, f(21)\}$ is

$$6 \cdot 70 \cdot 512 \equiv \boxed{40} \pmod{1000}.$$

RR 3 Part 2: Let $T = TNYWR$. In cyclic quadrilateral $ABCD$, $\angle BAD = 60^\circ$ and $BC = CD = T$. Suppose that AB is a positive integer. Find the median of all possible values of $2AB$.

Answer: $\boxed{81}$

Solution:



Let O be the center of the circle circumscribing quadrilateral $ABCD$. The radius of the circle is $T = 40$. As A ranges across major arc BD, AB increases from 0 (exclusive) to the length of the diameter ($2 \cdot 40 = 80$) then decreases to $BD = 40\sqrt{3}$ (exclusive). Since AB is an integer, the possible lengths are $\{1, 2, \ldots, 80\}$. So, the median of all possible values of $2AB$ is $\boxed{81}$.

RR 3 Part 3: Let $T = TNYWR$. Suppose $f(x)$ is a polynomial of degree 10, such that $f(i) = i$ for all $i = 1, 2, \ldots, 10$ and $f(11) = T$. Find the remainder when $f(13)$ is divided by 1000.

Answer: $\boxed{633}$

Solution: Let $g(x) = f(x) - x$. Since

$$f(i) = i \implies f(i) - i = g(i) = 0$$

for $i = 1, 2, \ldots, 10$, we have

$$g(x) = c(x-1)(x-2)\cdots(x-10),$$

for some constant c. From

$$f(11) = T = 81 \implies g(11) = f(11) - 11 = T - 11 = 81 - 11 = 70,$$

we have

$$g(11) = c(11-1)(11-2)\cdots(11-10) = 70 \implies c = \frac{70}{10!}.$$

We have

$$g(13) = \frac{70}{10!}(13-1)(13-2)\cdots(13-10) = \frac{70}{10!} \cdot \frac{12!}{1 \cdot 2} = 4620,$$

meaning

$$f(13) = g(13) + 13 = 4620 + 13 = 4633 \equiv \boxed{633} \pmod{1000}.$$

2022 SSMO Relay Round 4 Solutions

SMO Team

RR 4 Part 1: On each day, a robot finds a new organism with probability 0.7, an already discovered organism with probability 0.2, and nothing with probability 0.1. If a new organism is found, the robot correctly identifies it as new with probability 0.9, and if a discovered organism is found, the robot correctly identifies it as discovered with probability 0.75. The expected number of days until the robot reports that it has found a new organism can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{42}$

Solution: The probability the robot identifies an object as new on a given day is the sum of the probability it finds a new organism and correctly identifies it and the probability it finds a discovered organism and incorrectly identifies it. This is equal to

$$0.7 \cdot 0.9 + 0.2 \cdot (1 - 0.75) = 0.68.$$

Let the expected number of days be x. We have

$$x = 0.68 \cdot 1 + (1 - 0.68) \cdot (x + 1) \implies x = \frac{1}{0.68} = \frac{25}{17} \implies 25 + 17 = \boxed{42}.$$

RR 4 Part 2: Let $T = TNYWR$. The roots of $f(x) = x^3 + 5x + 8$ are r_1, r_2, r_3. Let $g_n(x)$ be a monic cubic polynomial with roots $r_1 + n, r_2 + n, r_3 + n$. Find the remainder when

$$\sum_{n=1}^{T} (-1)^n g_n(5)$$

is divided by 1000.

Answer: $\boxed{498}$

Solution: Since r_1, r_2, r_3 are the roots of $f(x)$, we have

$$f(x) = (x - r_1)(x - r_2)(x - r_3).$$

Since $r_1 + n, r_2 + n, r_3 + n$ are the roots of $g_n(x)$, we have

$$\begin{aligned} g_n(x) &= (x - (r_1 + n))(x - (r_2 + n))(x - (r_3 + n)) \implies \\ g_n(5) &= (5 - (r_1 + n))(5 - (r_2 + n))(5 - (r_3 + n)) \\ &= ((5 - n) - r_1)((5 - n) - r_2)((5 - n) - r_3) \\ &= f(5 - n) \\ &= (5 - n)^3 + 5(5 - n) + 8 \\ &= -n^3 + 15n^2 - 80n + 158. \end{aligned}$$

So,

$$\begin{aligned}\sum_{n=1}^{T}(-1)^n g_n(5) &= \sum_{n=1}^{42}(-1)^n f(5-n)\\ &= \sum_{i=1}^{21}\left((-1)^{2i-1}f(5-(2i-1))+(-1)^{2i}f(5-2i)\right)\\ &= \sum_{i=1}^{21}\left(f(5-2i)-f(6-2i)\right)\\ &= \sum_{i=1}^{21}\left(-12i^2+66i-96\right)\\ &= -12\left(\frac{21\cdot 22\cdot 43}{6}\right)+66\left(\frac{21\cdot 22}{2}\right)-96\cdot 21\\ &= -26502 \equiv \boxed{498} \pmod{1000}.\end{aligned}$$

RR 4 Part 3: Let $T = TNYWR$. Suppose $f : \mathbb{Z} \to \mathbb{Z}$ satisfies $f(1) = 1$, $f(2) = 12$, and

$$f(n+2) = 12f(n+1) - 20f(n)$$

for all positive integers n. Find the remainder when $f(T)$ is divided by 1000.

Answer: $\boxed{832}$

Solution: Consider the generating function

$$G(x) = \sum_{n\geq 1} f(n)x^n.$$

We have

$$\sum_{n\geq 1} f(n+1)x^n = \frac{\sum_{n\geq 2} f(n)x^n}{x} = \frac{G(x)-f(1)x}{x} = \frac{G(x)-x}{x}, \quad \text{and}$$
$$\sum_{n\geq 1} f(n+2)x^n = \frac{\sum_{n\geq 3} f(n)x^n}{x^2} = \frac{G(x)-f(1)x-f(2)x^2}{x^2} = \frac{G(x)-x-12x^2}{x^2}.$$

Multiplying the given equation by x^n and summing over $n \geq 1$, we have

$$\begin{aligned}\sum_{n\geq 1} f(n+2)x^n &= 12\sum_{n\geq 1} f(n+1)x^n - 20\sum_{n\geq 1} f(n)x^n \implies\\ \frac{G(x)-x-12x^2}{x^2} &= \frac{12G(x)-12x}{x} - 20G(x) \implies\\ G(x)-(x+12x^2) &= (12x)G(x)-(12x^2)-(20x^2)G(x) \implies\\ G(x) &= \frac{x}{20x^2-12x+1} = \frac{-\frac{1}{8}}{1-2x}+\frac{\frac{1}{8}}{1-10x}.\end{aligned}$$

This gives a coefficient extraction of

$$f(n) = [x^n]G(x) = -\frac{1}{8}2^n + \frac{1}{8}10^n.$$

So,

$$f(T) = f(498) = -\frac{1}{8}2^{498} + \frac{1}{8}10^{498} \equiv -2^{495} \pmod{1000}.$$

We have

$$2^{495} \equiv 2^{495-500} \equiv 32^{-1} \equiv 43 \pmod{125}.$$

Combining with $2^{495} \equiv 0 \pmod{8}$, we have $2^{495} \equiv 168 \pmod{1000} \implies f(498) \equiv -2^{495} \equiv \boxed{832} \pmod{1000}$.

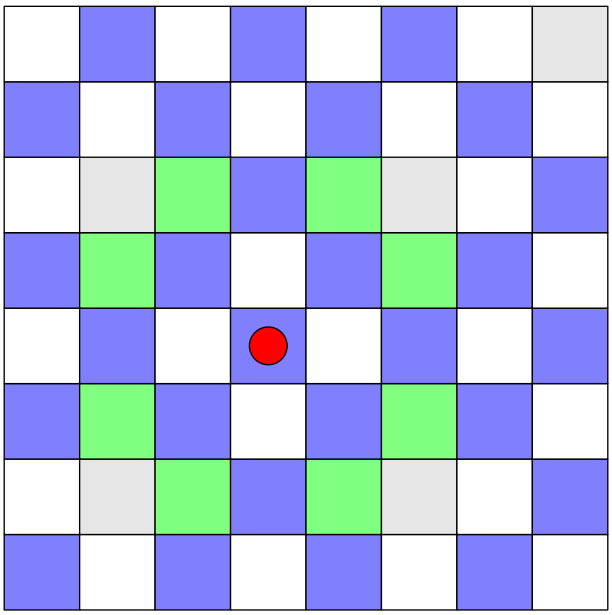
2022 SSMO Relay Round 5 Solutions

SMO Team

RR 5 Part 1: Consider an 8×8 chessboard with a knight in one of the center squares. The knight may move in an L-shaped fashion, going two squares in one direction and one square in a perpendicular direction, but cannot go outside the chessboard. How many squares can the knight reach in exactly two moves?

Answer: $\boxed{27}$

Solution:



The 8 green squares represent the squares the knight can reach in one move. The $\boxed{27}$ blue squares represent the squares the knight can reach in two moves.

RR 5 Part 2: Let $T = TNYWR$. Consider sequence $S = \{a_1, a_2, \ldots, a_{2022}\}$ of positive integers such that $a_1 \leq a_2 \leq \cdots \leq a_{2022}$ and $\text{lcm}(a_1, a_2, \ldots, a_{2022}) = 70T$. Assume $\gcd(a_i, a_j) = 1$ for all $1 \leq i < j \leq 2022$. Find the number of possible sequences S.

Answer: $\boxed{15}$

Solution: We have

$$70T = 70 \cdot 27 = 2 \cdot 3^3 \cdot 5 \cdot 7.$$

For each prime p, there must be exactly one a_i such that $p \mid a_i$. Since S is nondecreasing, it is determined by how the four prime powers are grouped into terms, so the number of sequences is the 4th Bell number, $\boxed{15}$.

RR 5 Part 3: Let $T = TNYWR$. Define $\text{cis}(x) = \cos(x) + i\sin(x)$ and $a_k = \text{cis}\left(\frac{k\pi}{T+1}\right)$. The value of

$$\sum_{k=1}^{T+1} \frac{|a_{2k} + a_{2k+2} - a_{2k-1+T}|}{|a_{2k+1} - (a_{2k} + a_{2k+2})|}$$

can be expressed as $a + b\cos(\frac{\pi}{c})$, where a, b, and c are positive integers. Find $a + b + c$.

Answer: $\boxed{64}$

Solution: Note that $\text{cis}(x) = e^{ix}$. Let $\theta = \frac{\pi}{T+1}$ and $\omega = e^{i\theta}$, meaning $\omega^{T+1} = e^{\pi i} = -1$. Each summand evaluates to

$$\begin{aligned}\frac{|a_{2k} + a_{2k+2} - a_{2k-1+T}|}{|a_{2k+1} - (a_{2k} + a_{2k+2})|} &= \frac{\left|\omega^{2k} + \omega^{2k+2} - \omega^{2k-1+T}\right|}{|\omega^{2k+1} - \omega^{2k} - \omega^{2k+2}|} \\ &= \frac{\left|\omega^{2k}\right| \left|1 + \omega^2 + \omega^{-2}\right|}{|\omega^{2k}| \, |\omega - 1 - \omega^2|}.\end{aligned}$$

Now,

$$\begin{aligned}\left|1 + \omega^2 + \omega^{-2}\right| &= |1 + (\cos(2\theta) + i\sin(2\theta)) + (\cos(-2\theta) + i\sin(-2\theta))| \\ &= |1 + 2\cos(2\theta)| = 1 + 2\cos(2\theta) \\ &= 1 + 2\left(2\cos^2(\theta) - 1\right) = (2\cos(\theta) - 1)(2\cos(\theta) + 1)\end{aligned}$$

and

$$\left|\omega - 1 - \omega^2\right| = \left|-\omega\left(\omega + \omega^{-1} - 1\right)\right| = |2\cos(\theta) - 1| = 2\cos(\theta) - 1,$$

which is positive since $\theta = \frac{\pi}{16} < \frac{\pi}{3}$. So, each summand evaluates to

$$\frac{\left|1 + \omega^2 + \omega^{-2}\right|}{|\omega - 1 - \omega^2|} = \frac{(2\cos(\theta) - 1)(2\cos(\theta) + 1)}{2\cos(\theta) - 1} = 2\cos(\theta) + 1.$$

So, the sum evaluates to

$$(T+1)\left(2\cos\left(\frac{\pi}{T+1}\right) + 1\right) = 32\cos\left(\frac{\pi}{16}\right) + 16 \implies 16 + 32 + 16 = \boxed{64}.$$